# Microeconomics I

Class 7 — The Slutsky equation
and the money value of a price change

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20 October 2026



## Plan for today

Warm-up; a mug, three prices

The Slutsky equation: substitution and wealth effects

Giffen goods, revisited

In pairs: gross and net cross effects

Break

The Slutsky matrix; the map of $x, v, h, e$

Compensating and equivalent variation

In pairs: a cheaper good

Deadweight loss; summary



### Slide 1 Title

**Preview version.** The full slides and handout are posted after class.

Class 7 closes the circle of classes 4 to 6. The two demands, $x$ from the budget and $h$ from the curve, are tied together by the Slutsky equation, which splits any price effect into a move along the indifference curve and a wealth effect; then the whole map of $x$, $v$, $h$ and $e$; and the payoff, what a price change is worth to the consumer in euros, measured at the old prices or at the new. MWG 3.G and 3.I.

### Slide 2 Plan for today

Before the break: last week's price rise taken apart into a substitution effect and a wealth effect; the Slutsky equation, which does the same split for a small change; what it says about market against compensated demand and about Giffen goods; and an exercise on class 6's gross-versus-net table. After the break: the Slutsky matrix from observable demand, the map of $x$, $v$, $h$, $e$, and the money value of a price change, compensating and equivalent variation, with a tax as the last example.

MWG 3.G and 3.I; Muñoz-Garcia chapter 2 is the gentler read.

## Practicalities

- Problem set 2: due at the start of class, Wed 21 Oct — on paper, one copy per group
- Office hours: Wednesdays 14:00–15:00, Room 445

  up to 20 minutes per student; those who book by e-mail have priority
- Slides and reading: pkocourek.com/teaching/micro-1 · pkocourek@ub.edu



### Slide 3 Practicalities

Problem set 2 is due at the start of class on Wednesday 21 October: the whole booklet, one per group, pages in order and stapled, all names in the box on the cover and nowhere else; extra sheets go at the end, each marked with the group and the part. Before you hand it in, photograph the code on the cover: once the booklet is marked, it opens your marked problem set. Today's material, the Slutsky equation and the money measures, is not on it.

Office hours are on Wednesdays, 14:00–15:00, Room 445, up to 20 minutes per student; booking by e-mail gives priority. Slides and reading: pkocourek.com/teaching/micro-1 · pkocourek@ub.edu.

## Warm-up: always true, or not?

Last class. True for every consumer? One minute with your neighbour.

| | |
|---|---|
| $p_1$ rises $\Rightarrow$ $h_1(p, u)$ does not rise | $p_1$ rises $\Rightarrow$ $x_1(p, w)$ does not rise |
| $e(p, u)$ is homogeneous of degree 0 in $p$ | $e\big(p, v(p, w)\big) = w$ |

Class 6, slides 10, 13, 14; class 5, slide 22. MWG Prop. 3.E.2, 3.E.4; (3.E.1).



### Slide 4 Warm-up: always true, or not?

Four one-line claims from last week, each about one of class 6's objects: compensated demand, market demand, the expenditure function, and the identity that ties $e$ to $v$. Before class, decide for each whether it holds for every consumer, and for the two in the top row say exactly what is held fixed in each.

Answers revealed in class. The point of the warm-up is the gap between the two lines of the top row, which the first hour of the class measures.

## A mug, three groups: whose prices should be equal?

**Simon Fraser University:** 77 students, a coffee mug, three groups at random

**Sellers** are given a mug: the lowest price at which they sell it

**Buyers**: the highest price they pay for one

**Choosers**: a mug or cash: the cash that is worth as much as the mug

### Which two medians should standard theory make equal?

buyers and choosers · choosers and sellers · buyers and sellers

A published experiment; the source in class.



**Slide 5** A mug, three groups: whose prices should be equal?

A published experiment with 77 students assigned at random to three groups: sellers, who were given a mug and named the lowest price at which they would part with it; buyers, who named the highest price they would pay for one; and choosers, who were offered a mug or cash and named the cash that was worth as much as the mug. Each group gives a median price, so one object is priced in three ways.

Before class, decide which two of the three medians a consumer as in our theory would make equal, and write down what each group gives up and what it receives. The source, and what the data said, come in class.

## A mug, three groups: what they said

# Revealed in class

Before class: which two groups face exactly the same choice?



**Slide 6** A mug, three groups: what they said

Revealed in class.

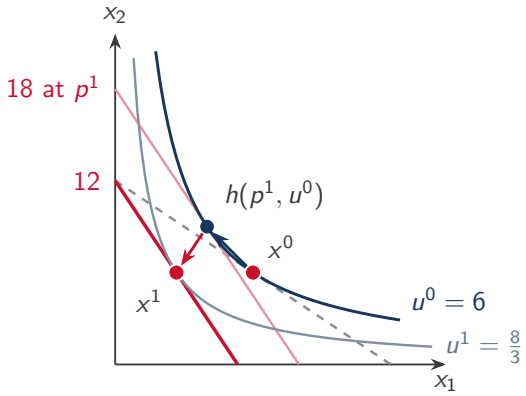
Before class: of the three groups, which two face exactly the same final choice? Then ask what, if anything, in our theory could make their prices differ.

# The Slutsky equation

MWG 3.G

slide 7 of 26

## Good 1 dearer: $x_1$ falls $3 \to 2$ along $u^0$, then $2 \to \frac{4}{3}$ from lost wealth




$x_1$: $3 \to 2 \to \frac{4}{3}$

substitution $-1$, wealth $-\frac{2}{3}$

$x_2$: $2 \to 3 \to 2$

substitution $+1$, wealth $-1$

18 euros at $p^1$ keep her on $u^0$

Substitution: along $u^0$, at the new prices.

Wealth: the parallel shift from 18 euros down to her 12.

MWG 3.G, pp. 71–72; class 6, slide 14. $u = x_1 x_2$, $w = 12$, $p^0 = (2, 3) \to p^1 = (4.5, 3)$.



**Slide 7** The Slutsky equation

Part 1: the Slutsky equation, which splits a price effect into a substitution effect and a wealth effect. MWG 3.G.

**Slide 8** Good 1 dearer: $x_1$ falls $3 \to 2$ along $u^0$, then $2 \to \frac{4}{3}$ from lost wealth

Class 6's price rise, good 1 from $2$ to $4.5$ with $12$ euros, moved her from $(3, 2)$ to $(\frac{4}{3}, 2)$. The slide splits that move at an intermediate bundle: the cheapest bundle on her old curve at the new prices, $h(p^1, u^0) = (2, 3)$, which $18$ euros would buy (class 6, slide 22). The first leg, along $u^0$, is the substitution effect: she reacts to the new relative price while held exactly as well off, so it always runs away from the dearer good. The second leg, from the $18$-euro line down to her actual $12$-euro line, is a parallel shift at the new prices, the wealth effect, which depends on the good; here both goods are normal, so both fall.

For good 2 the two legs cancel exactly because $x_1 x_2$ spends half of $w$ on each good whatever the prices: a feature of this consumer, not a rule. MWG compensates with utility held fixed, the Hicksian decomposition; the next slide writes the same split for a tiny price change. The trap is to draw the substitution step at the old prices: it is the new relative price along the old curve.

## Slutsky equation: compensated slope = market slope + wealth term

$$\frac{\partial h_\ell(p,u)}{\partial p_k} = \frac{\partial x_\ell(p,w)}{\partial p_k} + \frac{\partial x_\ell(p,w)}{\partial w}\, x_k(p,w) \quad \text{at } u = v(p,w)$$

why: differentiate $h_\ell(p,u) = x_\ell\big(p, e(p,u)\big)$ in $p_k$; $\partial e/\partial p_k = h_k = x_k$ (Shephard)

$$\frac{\partial x_\ell}{\partial p_\ell} = \underbrace{\frac{\partial h_\ell}{\partial p_\ell}}_{\leq 0:\ \text{substitution}} \underbrace{- x_\ell \frac{\partial x_\ell}{\partial w}}_{\text{wealth}}$$

$x_1x_2$ at $p = (2,3)$, $w = 12$: $-\frac{3}{2} = -\frac{3}{4} - 3 \cdot \frac{1}{4}$

A normal good falls more in market demand than in compensated demand.

MWG Prop. 3.G.3, p. 71, (3.G.3)–(3.G.4): $u$ continuous, $\succsim$ locally non-satiated and strictly convex.



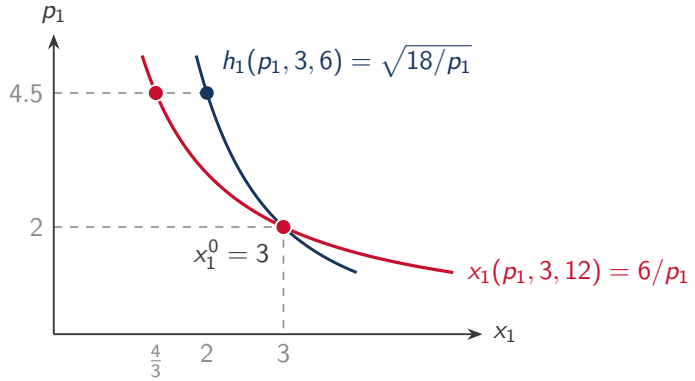
**Slide 9** Slutsky equation: compensated slope = market slope + wealth term

Read the first line as a sentence: the slope of compensated demand in a price equals the slope of market demand plus a wealth correction, the wealth slope times the quantity of the good whose price moved. The proof needs only class 6: $h$ is market demand at wealth $e(p,u)$, and when $p_k$ rises by a cent the cost of staying on the curve rises by $h_k$ cents (Shephard's lemma), which is $x_k$ at $u = v(p,w)$. So $x_k$ is how much poorer a one-cent rise in $p_k$ makes her, to first order.

The own-price form is slide 8 in derivatives: market effect = substitution effect, never positive, minus quantity times wealth slope. For $x_1x_2$ at $(2,3)$ and 12 the substitution part comes out as $-\frac{3}{4}$, class 6's compensated slope from its slide 20, the first check that the equation ties the two demands correctly; slide 8's $-1$ and $-\frac{2}{3}$ were the large-change version of the same two pieces. The slip is to put the wrong quantity in the wealth term: it is the good whose price moves, $x_k$, not the good whose demand we watch.

## Market demand falls faster than compensated: good 1 is normal

$u = x_1x_2$, $p_2 = 3$, $w = 12$, $u^0 = 6$: both curves pass through $x_1 = 3$ at $p_1 = 2$




At $p_1 = 4.5$: compensated demand 2, market demand $\frac{4}{3}$; the gap is the wealth effect.

MWG Figure 3.G.1(a), p. 71; text p. 72.

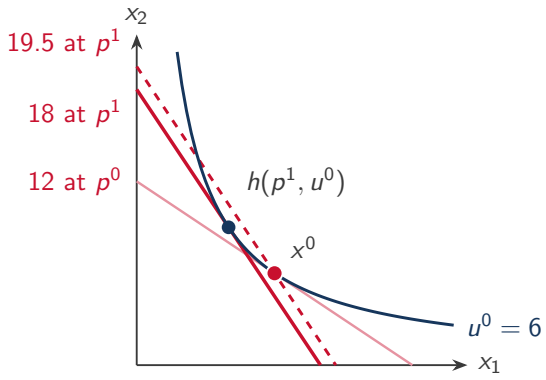


**Slide 10** Market demand falls faster than compensated: good 1 is normal

MWG draws demand with the price on the vertical axis, so a curve that reacts more to price looks flatter. The two curves share the point $x_1 = 3$ at $p_1 = 2$ because at that price twelve euros buy exactly $u = 6$, class 6's identity $e(p, v(p,w)) = w$. Away from that price they separate: above 2 a dearer good 1 makes her poorer and, good 1 being normal, she buys even less than compensated demand would, so the market curve lies to the left; below 2 a cheaper good makes her richer and the market curve lies to the right.

Why it matters: the whole gap between the curves is the wealth term of slide 9, and its sign is the sign of $\partial x_1/\partial w$. For an inferior good the two curves change places (MWG Figure 3.G.1(b)), and a Giffen good is the extreme case where the market curve bends the other way, slide 12. The trap is to say that the compensated curve is always the steeper one without saying for which goods.

## Two ways to compensate: keep $u^0$ (6 euros) or $x^0$ affordable (7.5)




Hicks: $e(p^1, u^0) - w = 18 - 12 = 6$
back on her old curve

Slutsky: $p^1 \cdot x^0 - w = 19.5 - 12 = 7.5$
her old bundle affordable again

Hicks $\leq$ Slutsky, equal to first order: so the compensated slopes of slide 9 carry Slutsky's name.

MWG 3.G, pp. 72–73, Figure 3.G.2; class 6, slide 22.

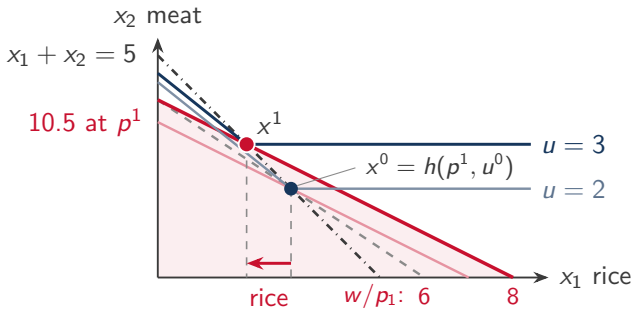


**Slide 11** Two ways to compensate: keep $u^0$ (6 euros) or $x^0$ affordable (7.5)

Two ways to make up for the price rise. Hicks gives her enough to get back to her old indifference curve, $18 - 12 = 6$ euros; Slutsky gives her enough to afford her old bundle again, $19.5 - 12 = 7.5$, class 6's fixed basket. Hicks can never cost more, because the old bundle is one way to reach $u^0$, so the cheapest way costs at most as much; with $19.5$ euros she would not keep $(3, 2)$ but substitute and end above $u^0$.

For a tiny price change the two agree to first order, since both equal $x^0 \cdot \Delta p$: the price gradient of $e$ at the start is $h(p^0, u^0) = x^0$ by Shephard's lemma. That is why the compensated slopes of slide 9, built with Hicksian compensation, carry Slutsky's name. The trap is to think the two compensations coincide for any size of change; they coincide only in the limit.

## Giffen needs inferior: class 5's rice, $3 \to 3 \to 2$




rice $p_1$: $2 \to 1.5$

$x_1$: $3 \to 3 \to 2$
substitution 0 (the kink on the food line $x_1 + x_2 = 5$), wealth $-1$

at $p_1 = 2$:
$$\frac{\partial x_1}{\partial p_1} = 0 - 3 \cdot (-1) = +3$$
$\partial h_1/\partial p_1 = 0$, $x_1 = 3$, $\partial x_1/\partial w = -1$

Substitution is never positive, so $\partial x_\ell/\partial p_\ell > 0$ needs $\partial x_\ell/\partial w < 0$: Giffen needs inferior.

Class 5, slides 19 and 22: $u = \min\{x_2,\ x_2 - 4(5 - x_1 - x_2)\}$, $w = 12$, $p_2 = 3$. MWG Prop. 3.G.3, p. 71.



**Slide 12** Giffen needs inferior: class 5's rice, $3 \to 3 \to 2$

Class 5's promise. The rice consumer bought less rice when it got cheaper, and the decomposition shows why that is possible. Her indifference curve has a kink at her old bundle, on the dash-dotted food line $x_1 + x_2 = 5$ where every one of her curves bends (class 5), and at the new price the compensating cost line still touches the curve only at the kink: held on her old curve she changes nothing, so the substitution effect is zero. The whole fall in rice is the wealth effect: cheaper rice leaves her $1.5$ euros to spare, and richer, she buys less rice, which is inferior for her (class 5, slide 19).

In derivatives, the own-price form of slide 9 gives the general lesson: the substitution term is never positive, so a positive market slope needs a negative wealth term, an inferior good whose wealth effect is strong enough to beat substitution. The converse fails: most inferior goods are not Giffen, because their substitution effect wins. Here $\partial h_1/\partial p_1 = 0$ is the weakest substitution effect possible, and that is what the kink buys.

## In pairs: net cross effect = gross + wealth term

Class 6, slide 21, at its start: $p = (2, 3)$, $w = 12$. $x_2$ from the budget.
$\sqrt{x_1} + \sqrt{x_2}$: the CES with $\rho = \frac{1}{2}$.

| $u$ | $x_1(p, w)$ | gross $\frac{\partial x_1}{\partial p_2}$ | wealth term $x_2 \frac{\partial x_1}{\partial w}$ | net $\frac{\partial h_1}{\partial p_2}$ |
|---|---|---|---|---|
| $\min\{x_1, x_2\}$ | $\frac{w}{p_1 + p_2}$ | | | |
| $x_1 x_2$ | $\frac{w}{2p_1}$ | | | |
| $\sqrt{x_1} + \sqrt{x_2}$ | $\frac{w\, p_2}{p_1(p_1 + p_2)}$ | | | |

Four minutes.

Answers revealed in class. Try it before class.



**Slide 13** In pairs: net cross effect = gross + wealth term

Class 6's slide 21 said that gross and net cross effects can disagree and that the gap is a wealth effect; this table measures the gap for the same three consumers at the start point. The recipe is the Slutsky equation with $\ell = 1$ and $k = 2$: differentiate each printed $x_1$ in $p_2$ and in $w$, find $x_2$ from the budget, and add the two pieces.

Try it before class, and before adding, decide which quantity multiplies the wealth slope and why. Answers revealed in class.

## (optional) The Slutsky equation in elasticities

$\varepsilon_{11} = (\partial x_1/\partial p_1)\, p_1/x_1$, $\varepsilon_{1w} = (\partial x_1/\partial w)\, w/x_1$, $\varepsilon^h_{11} = (\partial h_1/\partial p_1)\, p_1/h_1$;
budget share $b_1 = p_1 x_1/w$.

(1) Show: $\varepsilon_{11} = \varepsilon^h_{11} - b_1\, \varepsilon_{1w}$.

(2) $\varepsilon^h_{11} = -0.2$. For which $\varepsilon_{1w}$ is good 1 Giffen if $b_1 = 0.4$? If $b_1 = 0.02$?

At home if not done in class; the answers are in the slides released after class.

MWG (3.G.3), p. 71; elasticities as in MWG 2.E, p. 27; budget share $b_\ell$, p. 28. Invented numbers in (2).



**Slide 14** (optional) The Slutsky equation in elasticities

An optional exercise, to be done at home if class time runs out: slide 9's own-price form rewritten in elasticities, so that every term is unit-free. For (1), start from the Slutsky equation with $\ell = k = 1$ and ask what factor turns each slope into the elasticity defined on the slide; the budget share has to come out of the wealth term. For (2), recall from slide 12 which sign of $\varepsilon_{11}$ makes a good Giffen and solve for $\varepsilon_{1w}$.

The answers are in the slides and handout released after class. Before computing, say in words why the share of the budget spent on good 1 should matter for the wealth term at all.

# Break

Ten minutes



## The Slutsky matrix: class 6's $D_p h$, built from observable demand

$S(p, w)$: $s_{\ell k} = \dfrac{\partial x_\ell}{\partial p_k} + \dfrac{\partial x_\ell}{\partial w} x_k$

only Walrasian demand: observable in principle

$$S = \begin{pmatrix} -\frac{3}{2} + 3 \cdot \frac{1}{4} & 0 + 2 \cdot \frac{1}{4} \\ 0 + 3 \cdot \frac{1}{6} & -\frac{2}{3} + 2 \cdot \frac{1}{6} \end{pmatrix} = \begin{pmatrix} -\frac{3}{4} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{3} \end{pmatrix}$$

$= D_p h(p, 6)$: class 6, slide 20

negative semidefinite  symmetric  $S\,p = 0$

Demand whose $S$ is not symmetric or not negative semidefinite comes from no rational preferences.

MWG Prop. 3.G.2, pp. 69–70; 3.G, pp. 72–73: $S(p, w) = D_p h(p, u)$ (Prop. 3.G.3).





**Slide 15** Break

Ten minutes.

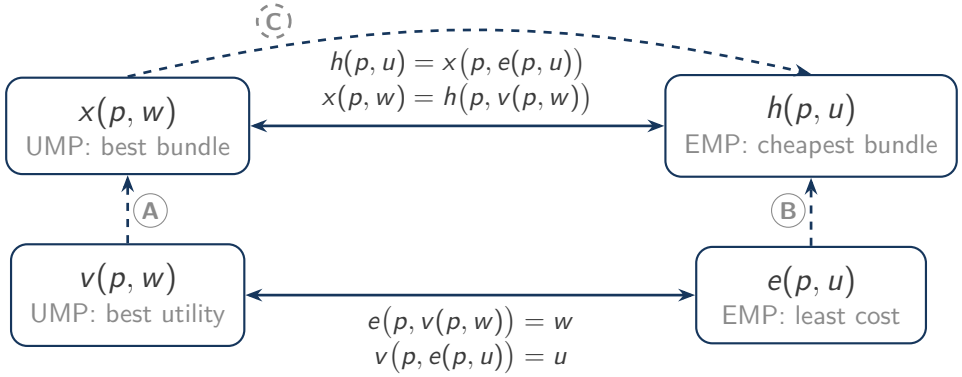
**Slide 16** The Slutsky matrix: class 6's $D_p h$, built from observable demand

Slide 9 rearranged: every compensated slope can be computed from market demand alone, its price slopes, its wealth slopes and the bundle, all observable in principle, while $h$ and $e$ never are. Doing it for $x_1 x_2$ returns exactly the matrix of class 6's slide 20, which there came from the second derivatives of $e$: class 6's promise kept. Because $S = D_p h = D_p^2 e$, it has three properties, each for its own reason: own effects never positive, because $e$ is concave in prices; equal cross effects, because the cross second derivatives of $e$ are equal; and $S\,p = 0$, because $h$ does not move when all prices scale.

Why it matters: these are restrictions on demand data that need no utility function to state. A demand whose $S$ fails symmetry cannot come from any preferences (MWG p. 73); that nobody would guess the symmetry from economics alone is Samuelson's remark. Gross effects have no such property, as class 5's rice and meat showed.

## In pairs: the map of $x$, $v$, $h$, $e$ — name the three derivative arrows

In pairs: name A, B, C — formula and name.




MWG Figure 3.G.3, p. 75 (a similar figure: Deaton and Muellbauer 1980); Props. 3.G.1, 3.G.3, 3.G.4.



**Slide 17** In pairs: the map of $x$, $v$, $h$, $e$ — name the three derivative arrows

MWG's map of the four functions of classes 4 to 7. The solid arrows are class 6's identities, which say that the two problems pick the same bundle and that $e$ and $v$ undo each other; the three dashed arrows are derivative results, one from class 5, one from class 6 and one from today.

Before class, name the three dashed arrows with their formulas, and for each say in which units the function at its tail is measured and whether the arrow delivers a function or only its slopes. Answers revealed in class.

# Money measures of welfare

MWG 3.I



**Slide 18** Money measures of welfare

Part 2: the money value of a price change, compensating and equivalent variation. MWG 3.I.

## Price the change in euros: EV at the old prices, CV at the new

money metric: $e\big(\bar{p}, v(p,w)\big)$

utility in euros at fixed prices $\bar{p}$

$\text{EV} = e(p^0, u^1) - w$

at the old prices, the change in wealth as good as the price change: $v(p^0, w + \text{EV}) = u^1$

$\text{CV} = w - e(p^1, u^0)$

at the new prices, what can be taken after the change to restore $u^0$: $v(p^1, w - \text{CV}) = u^0$

$u^0 = v(p^0, w), \quad u^1 = v(p^1, w)$

EV and CV: both negative when she loses

Both rank $p^0$ and $p^1$ correctly; their euros differ.

MWG 3.I, pp. 81–82: Figure 3.I.1; (3.I.1), (3.I.2), after Hicks (1939).

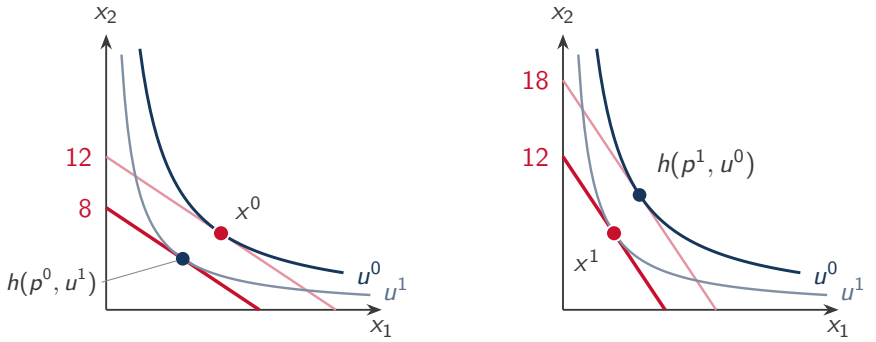


**Slide 19** Price the change in euros: EV at the old prices, CV at the new

Utility numbers mean nothing by themselves (class 3); euros do. The money metric fixes a price vector $\bar{p}$ and asks how much money she would need at $\bar{p}$ to be as well off as she is at $(p, w)$. It is a utility function for the same preferences, increasing in $v$, and it does not change when $u$ is relabelled, because $e$ sees only the curve (class 6, slide 12). Two natural choices of $\bar{p}$ give Hicks's two measures: the old prices for EV, the new prices for CV.

EV answers: what change in wealth, with no change in prices, would do to her exactly what the price change does. CV answers: after the change, how much could be taken from her to bring her back to her old utility, a negative number meaning she must be given money. Both carry the sign of the welfare change, so both rank the two price vectors correctly; their sizes differ because they price utility at different prices. The trap is to swap the prices and the utility levels inside $e$: EV uses the old prices with the new utility, CV the new prices with the old utility.

## Good 1 to 4.5: EV $= -4$ euros, CV $= -6$ euros




**EV: old prices**

$e(p^0, u^1) - w = 8 - 12 = -4$

**CV: new prices**

$w - e(p^1, u^0) = 12 - 18 = -6$

Old prices value her loss at 4 euros, new prices at 6: EV $>$ CV, good 1 is normal.

MWG Figure 3.I.2, p. 82; $u = x_1 x_2$, $w = 12$, $p^0 = (2, 3)$, $p^1 = (4.5, 3)$.

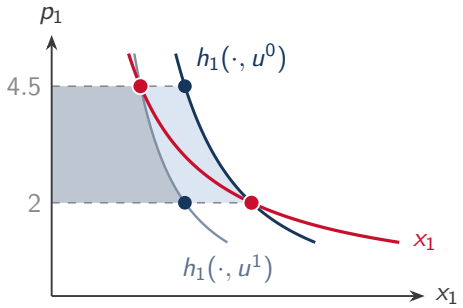


**Slide 20** Good 1 to 4.5: EV $= -4$ euros, CV $= -6$ euros

The same event, class 6's price rise, priced twice. Left, at the old prices: the utility she ends up with, $u^1$, could have been reached at $p^0$ for $8$ euros, so the price rise did her as much harm as losing $4$ of her $12$. Right, at the new prices: getting back to $u^0$ costs $18$ (class 6), $6$ more than she has. Both numbers are negative in MWG's convention because she loses.

Why they differ: good 1 is normal, and valuing her loss at the old, cheap price of good 1 makes the loss look smaller, so EV $>$ CV; for an inferior good the order reverses (MWG p. 83). The fixed basket of class 6 said she needs $7.5$ more; the truth, CV, is $6$: concavity of $e$ makes the fixed-basket number an upper bound on the loss. MWG's Figure 3.I.2 reads the euros on the vertical axis because it sets $p_2 = 1$; here $p_2 = 3$, so read the cost labels on the lines.

## $-\mathbf{CV} = 6$ and $-\mathbf{EV} = 4$ are areas left of Hicksian demand




$-\text{CV}$: both shades  $-\text{EV}$: dark

$$-\text{CV} = \int_2^{4.5} h_1(p_1, 3, u^0)\, dp_1 = 6$$

$$-\text{EV} = \int_2^{4.5} h_1(p_1, 3, u^1)\, dp_1 = 4$$

$$\int_2^{4.5} x_1(p_1, 3, 12)\, dp_1 = 6 \ln 2.25 \approx 4.87$$

market demand: in between

Shephard turns a change in $e$ into an area under compensated demand.

MWG 3.I, pp. 83 and 89: (3.I.3), (3.I.4), (3.I.8); Figure 3.I.3.



**Slide 21** $-\text{CV} = 6$ and $-\text{EV} = 4$ are areas left of Hicksian demand

Why areas: by Shephard's lemma $\partial e/\partial p_1 = h_1$, so the change in $e$ as $p_1$ moves from $2$ to $4.5$ at a fixed utility is the integral of compensated demand between the two prices, the area to the left of that Hicksian curve. CV holds utility at $u^0$ and uses the outer curve; EV holds it at $u^1$ and uses the inner one. The market demand curve starts on the $u^0$ curve, where $12$ euros buy $u^0$, and ends on the $u^1$ curve, where they buy $u^1$, so for a normal good its area lies between the two: MWG's area variation, which overstates one measure and understates the other.

When do all three agree? If good 1 has no wealth effect, for instance preferences quasilinear in the other good, the three curves coincide and CV = EV = the change in Marshall's consumer surplus. The trap is to take the market-demand area as the exact welfare change; it is exact only without a wealth effect, and slide 24 shows what the wrong curve would do for a tax.

## CV, EV and the mugs

# Revealed in class

Before class: which group's price is a CV of getting a mug, which an EV?



**Slide 22** CV, EV and the mugs

Revealed in class.

Before class: with the definitions of slide 19, decide which group's price is a compensating variation of getting a mug and which an equivalent variation, and which two groups theory should therefore make equal.

## In pairs: good 1 falls to 0.5: what is the gain worth?

$u = x_1 x_2$, $w = 12$, $p^0 = (2, 3) \to p^1 = (0.5, 3)$.

| question | answer |
|---|---|
| new bundle $x^1$ and utility $u^1$ | |
| $h(p^1, u^0)$: split $x_1$ into substitution and wealth | |
| $\text{CV} = w - e(p^1, u^0)$ | |
| $\text{EV} = e(p^0, u^1) - w$ | |

Five minutes.

Answers revealed in class. Try it before class.

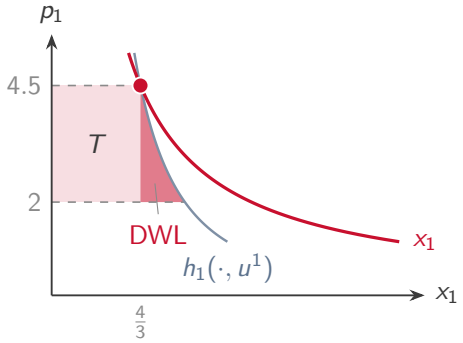


**Slide 23** In pairs: good 1 falls to 0.5: what is the gain worth?

The same consumer as all day, now with good 1 getting cheaper, so every number should come out as a gain. Try the four lines before class: the new choice, the compensated bundle on the old curve at the new prices with the price effect split into its two steps, and then the two money measures from slide 19's definitions.

Before computing, write down which prices and which utility level go into $e$ for each measure, and predict which of the two will be larger and why. Answers revealed in class.

## Tax good 1 by 2.5: it raises 3.33, costs her 4: deadweight loss 0.67




tax $t = 2.5$ per unit: $p_1$: $2 \to 4.5$

revenue $T = t\, x_1(p^1, w) = 2.5 \cdot \frac{4}{3} \approx 3.33$

her loss $-\text{EV} = 4$

deadweight loss $= 4 - 3.33 \approx 0.67$

The tax takes more from her than it brings in: a triangle left of Hicksian demand.

MWG Example 3.I.1, pp. 84–85, (3.I.5), Figure 3.I.4(a); own numbers.

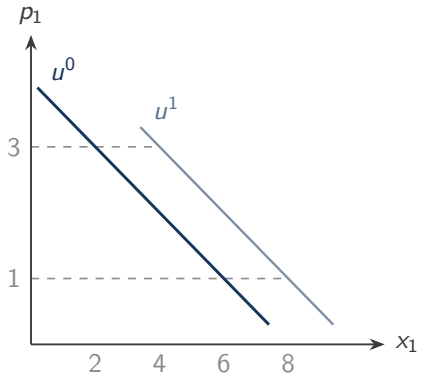


**Slide 24** Tax good 1 by 2.5: it raises 3.33, costs her 4: deadweight loss 0.67

Read the price rise as a per-unit tax of $2.5$ on good 1. The state collects the tax on what she buys after the change. Her loss, measured as EV, is larger than that revenue: if the state took the same sum as a lump sum instead, prices would stay at $p^0$ and she would keep more than the $8$ euros that her post-tax utility costs at $p^0$, so she would be better off and the state no poorer. The gap is the deadweight loss of the tax, MWG's Example 3.I.1.

In the figure the EV area of slide 21 splits into the revenue rectangle and the sliver beside it. The sliver is never negative, because compensated demand never rises in its own price, and it is zero only when she cannot substitute, as for Leontief, where it has no width. MWG measures it against the Hicksian curve at $u^1$, not the market curve; the market curve would give the wrong triangle, which is the trap.

## (optional) CV and EV from two compensated demand curves




$p_1$: $1 \to 3$; $p_2$, $w$ fixed.

$h_1(p_1, u^0) = 8 - 2p_1$, $h_1(p_1, u^1) = 10 - 2p_1$

(1) CV and EV?

(2) Normal or inferior?
Is EV $>$ CV, as on slide 20?

At home if not done in class; the answers are in the slides released after class.

MWG 3.I, p. 83: (3.I.3), (3.I.4), Figure 3.I.3. Own numbers.



**Slide 25** (optional) CV and EV from two compensated demand curves

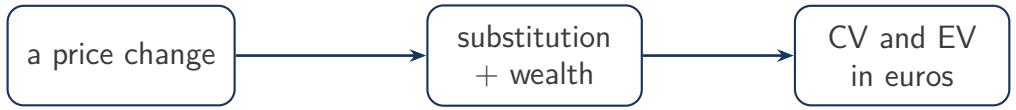
An optional exercise, to be done at home if class time runs out: no utility function this time, only two straight compensated demand curves for good 1, one at the old utility and one at the new, and a price rise from 1 to 3. For (1), use slide 21: each measure is the area to the left of one of the lines between the two prices, so decide which line belongs to which measure and what sign a loss carries. For (2), ask what the position of the $u^1$ line relative to the $u^0$ line says about how her demand for good 1 responds to wealth, and let that settle the order of the two measures.

The answers are in the slides and handout released after class. Before computing, write down which of $u^0$ and $u^1$ is the higher utility here, and why.

## Today in one line

Slutsky:
$\partial h/\partial p = \partial x/\partial p + (\partial x/\partial w)\, x^\top$

$e$ at $p^1$ or $p^0$:
areas left of $h$




Wednesday: Uncertainty I: lotteries, the independence axiom, the expected utility theorem and the uniqueness of the vNM utility function.
Money lotteries, risk aversion, certainty equivalent, risk premium.
Problem set 2: due at the start of class, Wednesday 21 October — on paper, one copy per group.

Read MWG 6.B–6.C (or Muñoz-Garcia ch. 5) for Wednesday 21 October.



**Slide 26** Today in one line

In one line: a price change moves her along her curve, the substitution effect, never towards the dearer good, and shifts her budget, the wealth effect, whose sign depends on the good; that is the Slutsky equation, and it is why a Giffen good must be inferior. The matrix of substitution effects can be computed from market demand and must be symmetric and negative semidefinite. The map ties $x$, $v$, $h$ and $e$ together. In euros a price change is worth CV at the new prices or EV at the old, areas to the left of Hicksian demand, equal when there is no wealth effect.

Wednesday 21 October: uncertainty, lotteries, the independence axiom, the expected utility theorem and the uniqueness of the vNM utility function; money lotteries, risk aversion, certainty equivalent, risk premium. Reading: MWG 6.B–6.C, or Muñoz-Garcia chapter 5.

Problem set 2 is due at the start of class on Wednesday 21 October, on paper, one copy per group.